

**THIRD SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2021**

Mathematics

MAT 3B 03—CALCULUS AND ANALYTIC GEOMETRY

(2014—2018 Admissions)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 3B 03—CALCULUS AND ANALYTIC GEOMETRY

(Multiple Choice Questions for SDE Candidates)

1. $\int \tan x \, dx = \underline{\hspace{2cm}}.$

(A) $\ln |\cos x| + c.$

(B) $\ln |\sec x| + c.$

(C) $\sec^2 x + c.$

(D) $\ln |\sin x| + c.$

2. $\int a^x \, dx = \underline{\hspace{2cm}}.$

(A) $\frac{a^x}{\ln a}.$

(B) $a^x \ln a.$

(C) $\frac{\ln a}{a^x}.$

(D) $\frac{a^{x+1}}{x+1}.$

3. $\coth^{-1} x = \underline{\hspace{2cm}}.$

(A) $\frac{1}{\tanh x}.$

(B) $\tanh^{-1} \left(\frac{1}{x} \right).$

(C) $\coth^{-1} \left(\frac{1}{x} \right)$

(D) $\frac{1}{\tanh^{-1} \left(\frac{1}{x} \right)}.$

4. $\lim_{x \rightarrow 0} \frac{\sin x}{x^2} = \underline{\hspace{2cm}}.$

(A) $-1.$

(B) $0.$

(C) $\infty.$

(D) $-\infty.$

5. Range of $\tanh x$ is $\underline{\hspace{2cm}}.$

(A) $[-1, 1].$

(B) $(-1, 1).$

(C) $(-\infty, \infty).$

(D) $(0, \infty).$

6. $\lim_{n \rightarrow \infty} \frac{2n}{n+1} = \underline{\hspace{2cm}}.$

(A) $0.$

(B) $\frac{2}{3}.$

(C) $2.$

(D) $\infty.$

7. $\lim_{n \rightarrow \infty} x^{1/n} (x > 0) = \underline{\hspace{2cm}}.$

- (A) 0. (B) 1.
(C) ∞ . (D) None of these.

8. Fourth term of the sequence $\left\{ \frac{(-1)^{n+1}}{2^{n+1}} \right\}$ is:

- (A) $\frac{1}{2^5}$. (B) $-\frac{1}{2^5}$.
(C) $\frac{1}{2^4}$. (D) $-\frac{1}{2^4}$.

9. The radius of convergence of the power series $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$ is $\underline{\hspace{2cm}}.$

- (A) 0. (B) 1.
(C) 2. (D) ∞ .

10. Co-efficient of x^2 in the Maclaurin's series expansion of $f(x) = \ln(x+1)$ is:

- (A) -1. (B) $-\frac{1}{2}$.
(C) $\frac{1}{2}$. (D) 1.

11. Equation of the directrix of the parabola $y^2 = 8x$ is:

- (A) $x = 2$. (B) $y = 2$.
(C) $x = -2$. (D) $y = -2$.

12. $x^2 + 2xy + y^2 + 2x - y + 2 = 0$ represent:

- (A) Parabola. (B) Ellipse.
(C) Hyperbola. (D) Circle.

13. Find $\frac{d^2y}{dx^2}$ if $x = \cos t, y = \sin t$:

- (A) $\operatorname{cosec}^2 t$. (B) $-\operatorname{cosec}^2 t$.
(C) $\operatorname{cosec}^3 t$. (D) $-\operatorname{cosec}^3 t$.

14. The equation $r^2 = \sin 2\theta$ is symmetric about :

- (A) x -axis. (B) y -axis.
(C) Origin. (D) The line $\theta = \frac{\pi}{4}$.

15. $\sum (-1)^n \frac{1}{n}$ is _____.

- (A) Converges. (B) Diverges.
(C) Absolutely convergent. (D) None.

16. $\lim_{x \rightarrow \infty} \left(\frac{1}{n^2} \right)^{\frac{1}{n}} =$ _____.

- (A) 0. (B) 1.
(C) ∞ . (D) None.

17. If $f(x) = 5x - 4$, if $x < 2$ $\lim_{x \rightarrow 2} f(x) =$ _____.
 $2(x^2 - 1)$, if $x > 2$

- (A) 0. (B) 6.
(C) 14. (D) Does not exist.

18. $y = \sin(\sin x)$. Then $\frac{dy}{dx} - \cos(\sin x) \cos x =$ _____.

- (A) 0. (B) $\cos x$.
(C) $\sin x$. (D) 1.

19. The function $f(x) = |x|$ is _____.

- (A) Continuous at $x = 0$. (B) Discontinuous at $x = 0$.
(C) Differentiable at $x = 0$. (D) Not differentiable at $x = 0$.

20. The series $\sum n^m x^n$ is converged if _____.

- (A) $x > 1$ and $x = 1$ when $m < -1$. (B) $x > 1$ and $x = 1$ when $m > -1$.
(C) $x < 1$ and $x = 1$ when $m < -1$. (D) $x < 1$ and $x = 1$ when $m > -1$.

**THIRD SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2021**

Mathematics

MAT 3B 03—CALCULUS AND ANALYTIC GEOMETRY

(2014—2018 Admissions)

Time : Three Hours

Maximum : 80 Marks

Part A (Objective Type Questions)

*Answer all **twelve** questions.*

Each question carries 1 mark.

1. Find the eccentricity of the hyperbola $9x^2 - 16y^2 = 144$.
2. Find $\lim_{n \rightarrow \infty} \frac{4 - 7n^6}{n^6 + 3}$.
3. The inverse function of $\ln(x)$, $x > 0$ is
4. If $f(x) = x^3 - 2$, find the value of $\frac{df^{-1}}{dx}$ at $x = f(2) = 6$.
5. Find a formula for the n^{th} term of the sequence 1, -4, 9, -16, 25,
6. The centre-to-focus distance in the case of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is
7. Write a parametrization of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.
8. $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = \dots\dots\dots$

Turn over

9. Give an example of a conditionally convergence sequence.
10. Define a harmonic series.
11. If $\log_a x = \frac{\ln(x)}{k}$, then $k =$
12. The polar equation of the circle $x^2 + y^2 = 25$ is

(12 × 1 = 12 marks)

Part B (Short Answer Type)

*Answer any **nine** questions.
Each question carries 2 marks.*

13. Find the focus of the parabola $x^2 = 100y$.
14. Find $\lim_{x \rightarrow 0^+} x \cot x$.
15. Find $\lim_{n \rightarrow \infty} \frac{\cos n}{n}$.
16. State non-decreasing sequence theorem.
17. For what values of x do the power series $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$ converge ?
18. Evaluate $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^2}$.
19. Determine the conic section from the equation $3x^2 - 6xy + 3y^2 + 2x - 7 = 0$.
20. Identify the geometric figure in the cartesian plane represented by the polar equation $r \cos(\theta - \pi/3) = 2$.
21. Test the convergence of $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$.

22. For what values of x do $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}$ converge?

23. Write the formula for finding the area using polar co-ordinates.

24. Find the polar equation of the hyperbola with eccentricity $3/2$ and directrix at $x = 2$.

(9 × 2 = 18 marks)

Part C (Short Essay Type)

Answer any **six** questions.

Each question carries 5 marks.

25. Using the definition of natural logarithm, prove that $\ln(xy) = \ln x + \ln y$.

26. Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right)$.

27. Find a formula for the n^{th} partial sum of the series $2 + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \dots + \frac{2}{3^n - 1}$ and use it to find the series's sum if the series converges.

28. Use integral test to show that $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges for $p > 1$.

29. Show that the point $(2, 3\pi/4)$ lies on the curve $r = 2 \sin 2\theta$.

30. Find the Maclaurin series for the function $f(x) = \sinh x$.

31. Find the radius and interval of convergence of the series $\sum_{n=0}^{\infty} (-1)^n (4x + 1)^n$.

32. Calculate the value of e with an error less than 10^{-6} .

33. Draw the curve represented by the parametric equations $x = \cos t, y = -\sin t, t \in [0, \pi]$.

(6 × 5 = 30 marks)

Turn over

Part D (Essay Type)

Answer any **two** questions.

Each question carries 10 marks.

34. Find the area of the surface generated by revolving the right handed loop of the lemniscate $r^2 = \cos 2\theta$ about y-axis.
35. Find the Taylor series generated by $f(x) = 1/x$ at $a = 2$. Where if anywhere, does the series converges to $1/x$?
36. a) The x and y axes are rotated through an angle of $\pi/4$ radians about the origin. Find the converted equation of the hyperbola $2xy = 9$ in the new co-ordinates.
- b) Evaluate $\lim_{x \rightarrow \left(\frac{\pi}{2}\right)^-} \frac{\sec x}{1 + \tan x}$.

(2 × 10 = 20 marks)

**THIRD SEMESTER (CUCBCSS—UG) [SPECIAL] DEGREE EXAMINATION
NOVEMBER 2019**

Mathematics

MAT 3B 03—CALCULUS AND ANALYTIC GEOMETRY

Time : Three Hours

Maximum : 80 Marks

Part A

Answer all the twelve questions.

Each question carries 1 mark.

1. Write the parametric equations of the circle $x^2 + y^2 = 1$.
2. Find the Taylor series for $f(x) = e^x$ at $x = 0$.
3. Find the foci of the hyperbola $\frac{y^2}{4} - \frac{x^2}{5} = 1$.
4. Evaluate $\int_0^{\pi} \frac{\sin t}{2 - \cos t} dt$.
5. Examine whether $xy - y^2 - 5y + 1 = 0$ represents a parabola, ellipse or hyperbola.
6. Prove that $e^{x + \ln 2} = 2e^x$.
7. Evaluate $\frac{d}{dt} \left(\tanh \sqrt{1 + t^2} \right)$.
8. Examine whether $\sum_{n=1}^{\infty} \frac{n+1}{n}$ converges or diverges.
9. Find the Taylor polynomial of order zero generated by $f(x) = \frac{1}{x}$ at $a = 2$.

Turn over

10. Find the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

11. State Leibniz's Theorem for an alternating series.

12. Evaluate $\int_0^{\pi/2} e^{\sin x} \cos x \, dx$.

(12 × 1 = 12 marks)

Part B

Answer any **nine** questions.

Each question carries 2 marks.

13. Find $\frac{dy}{dx}$ if $y = x^x$, $x > 0$.

14. Prove that the alternating series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \text{converges.}$$

15. Examine whether $x^2 + xy + y^2 - 1 = 0$ represents a parabola, ellipse or hyperbola.

16. Show that $\ln x$ grows slower than x as $x \rightarrow \infty$.

17. Evaluate $\int_{-\pi/2}^{\pi/2} \frac{4 \cos \theta}{3 + 2 \sin \theta} d\theta$.

18. Find the Taylor series for $f(x) = e^x$ at $x = 0$.

19. Graph the set of points whose polar co-ordinates satisfy the conditions $1 \leq r \leq 2$ and $0 \leq \theta \leq \pi/2$.

20. For what values of x do the series $\sum_{n=0}^{\infty} n! x^n$ converges.

21. Evaluate $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$.
22. Define absolute convergence.
23. Examine whether $\sum_{n=1}^{\infty} (-1)^{n+1}$ converges or diverges.
24. Find y if $\ln y = 3t + 5$.

(9 × 2 = 18 marks)

Part C

*Answer any six questions.
Each question carries 5 marks.*

25. Find the directrix of the parabola $r = \frac{25}{10 + 10 \cos \theta}$.
26. Find the Maclaurin's series for $f(x) = \sin 3x$.
27. Find $\lim_{x \rightarrow \infty} x^{\frac{1}{x}}$.
28. Graph the curve $r = 1 - \cos \theta$.
29. Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right)$.
30. Find the tangent to the right-hand hyperbola branch $x = \sec t$, $y = \tan t$, $-\frac{\pi}{2} < t < \pi/2$ at the point $(\sqrt{2}, 1)$ where $t = \frac{\pi}{4}$.

31. Using Integral test show that the p -series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \dots + \frac{1}{n^p} + \dots \text{ converges if } p > 1 \text{ and diverges if } p \leq 1.$$

32. Find the centroid of the first quadrant of the astroid $x = \cos^3 t$, $y = \sin^3 t$, $0 \leq t \leq 2\pi$.

33. Find the length of the Cardioid $r = 1 - \cos \theta$.

(6 × 5 = 30 marks)

Part D

Answer any two questions.

Each question carries 10 marks.

34. Find the length of the astroid $x = \cos^3 t$, $y = \sin^3 t$, $0 \leq t \leq 2\pi$.

35. Show that the Maclaurin's series for $\sin x$ converges to $\sin x$ for all x .

36. Find the sum of the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

(2 × 10 = 20 marks)